Heavy metals in baby wipes

HEAVY METAL INDEX · SYNTHESIS REPORT

September 2026

Karen Pendergrass · Heavy Metal Index, Institute of Contaminant Standards

DOI 10.83376/hmi.heavy-metals-in-baby-wipes

CITE AS

Pendergrass, K. (2026). Heavy metals in baby wipes. Heavy Metal Index. https://doi.org/10.83376/hmi.heavy-metals-in-baby-wipes

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

Pb, tAs, iAs, tHg, MeHg, Cd, Cr, Cr-VI, Ni, Sn, Al, Sb, heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

First published 2026-09-25. This revision 2026-09-26.

Abstract

Four datasets have measured heavy metals in baby wipes, each on a different analytical basis, and antimony is the only metal detected in all of them. Tin, aluminum, inorganic arsenic, methylmercury, and hexavalent chromium have not been measured in a wipe, and no study has measured how much of any metal crosses infant skin, so a concentration in a wipe is not yet a dose.

1 | 1. Introduction: a wipe is three objects, measured three ways

A baby wipe is a nonwoven sheet, a liquid that wets it, and a package that holds both. Each can carry a metal by a different route: the sheet through the chemistry used to make its fiber, the liquid through its water and ingredients, and the package through its own materials. The question a caregiver or a regulator eventually asks, how much of a metal reaches an infant from a wipe, depends on four quantities: the metal content of the sheet and the lotion on a mass basis, the amount of product that touches skin, the time it stays there, and the fraction that is absorbed. The literature reviewed here supplies part of the first quantity and none of the other three.

That imbalance is the central finding of this synthesis. The wipe record is recent, small, and methodologically divided, and its datasets were produced by laboratories that answered different questions with different preparations. A concentration printed in one of those datasets is therefore not interchangeable with a concentration printed in another, nor with the unit in which a cosmetic or textile limit is written. The unit and preparation behind each number are treated here as part of the finding, not as a technicality to be converted away.

The analyte panel covers lead, arsenic, mercury, cadmium, chromium, nickel, tin, and aluminum, with speciation kept wherever it changes the hazard. Antimony is added as a ninth metal because it is the metal the wipe measurements actually find.

Review approach and evidentiary boundaries

 © All rights reserved

This is a targeted narrative synthesis, not a systematic review or meta-analysis. It draws on the documents held in the Heavy Metal Index corpus for baby wipes as of 25 September 2026: four datasets that measure metals in wipes or wipe components, two diaper studies used only as adjacent-product context, a polyester-textile release study, a hazard monograph, a pediatric patch-test meta-analysis, a manufacturing description of wipe lotion, manufacturer filings to United States state reporting systems, and the regulatory instruments discussed below. Where a document was available only as a publisher abstract, an annex image, or a secondary reproduction, the text says so at the point of use. A statement that a metal has not been measured in a baby wipe refers to the literature identified for this synthesis; it is a statement about the record, not about the product.

Values are reported in the unit and on the basis the source printed. The datasets are not pooled, no market detection rate is computed, and no value is converted to a content basis except in one labeled reconstruction that is not treated as a result. Products are identified only by the codes their authors assigned. Brands are not named, and nothing here ranks products now on sale.

2 | 2. Product anatomy: sheet, liquid, and package

The sheet is a nonwoven. Fibers named on products and in the literature include rayon, a regenerated cellulose usually made from wood pulp; polyester, including polyethylene terephthalate (PET); cotton; polypropylene; and blends of these. In the Korean digest study, 12 wet-wipe sheets were labeled rayon plus polyester and 5 were labeled rayon plus PET, while the rest carried labels of rayon alone, cotton, rayon with polyester and cotton, pulp with polypropylene, or no fiber statement.[2] Those labels are what the authors printed for each coded product. They are not a fiber analysis.

The liquid wets the sheet, cleans the skin, and carries preservatives and, in scented products, fragrance. Cheriaa and Boubaker describe one lotion for polyester and viscose baby wipes as purified water, a surfactant, phenoxyethanol, and EDTA, and they cite a recommendation that baby wipes be buffered toward skin pH.[5] Their paper names no metal salt and reports no metal concentration, so it describes what a lotion contains by design rather than what metals it carries. The same liquid is where the non-metal constituents of a wipe sit, and those constituents are a separate question from the one addressed here.

Manufacture is where antimony enters the polyester account. PET is made by polymerizing ethylene glycol with terephthalic acid, and a metal catalyst speeds that reaction; antimony trioxide is the historical catalyst, with titanium, germanium, and other metals used as alternatives.[2] Rayon is produced by a cellulose process, not a PET polymerization, so the catalyst chemistry predicts antimony in polyester fiber and does not by itself predict it in rayon.

The package is a separate article. The Washington study measured metals in some wipe packages as well as in wipe components, and a package result is evidence about the package.[1] No study reviewed here shows that a metal in a package moved into the sheet or the lotion.

3 | 3. The measured record

Four datasets measure metals in baby wipes or their parts, and they answer four different questions. The Washington results are parts per million in a component that had already screened positive by X-ray fluorescence. The Korean results are micrograms per liter in a digest of the sheet. The Chinese results are a per-liter antimony reading from an annex image, with no described preparation. The Turkish results are parts per billion in the solution that entered the instrument, with the squeezed liquid, the digested fiber, and the digested

packaging kept apart. Table 1 sets them side by side without converting any of them. Two diaper studies follow as adjacent-product context, clearly separated from the wipe results.

3.1 Washington State: screened components in parts per million

The Washington State Department of Ecology report, first published in 2014 and revised in 2021 to update contact details, screened children's and consumer products by X-ray fluorescence and sent components in which the screen found a metal of concern to a contract laboratory, which used inductively coupled plasma mass spectrometry (ICP-MS) and cold-vapor atomic absorption for mercury.[1] The design matters for interpretation: these rows are a screened list, not a sample of wipes on the market.

Three wipe components were reported at 40.6, 61.3, and 70.6 ppm antimony, with lead at 0.4, 0.8, and 1.4 ppm and total chromium at 0.5 to 0.6 ppm. Arsenic, cadmium, and mercury are printed as 0.0 in those rows.[1] The report does not identify the fiber, does not say whether the component was the sheet, the lotion, or both, and has no column for nickel, tin, or aluminum. It also states that hexavalent chromium analysis was outside the study, so the chromium figures are total chromium.

The results report does not describe the digestion used on the wipe rows. The methods sentence cites the four 2012 quality-assurance project plans behind the study, and each of those plans specifies microwave digestion by EPA 3052 with measurement by ICP-MS, or cold-vapor atomic absorption for mercury, with a reporting limit of 1.0 ppm for antimony, arsenic, cadmium, and lead and 0.1 ppm for mercury.[6,7,8,9] Chromium at 1.0 ppm appears only in Publications 12-07-023 and 12-07-024, and nickel appears in none of them.[8,9] The plans record the intended method design; they do not state that a particular wipe row was digested that way. Because the results report does not apply the plans' reporting limits to the wipe rows, lead at 0.4 and 0.8 ppm and chromium at 0.5 and 0.6 ppm are left as printed here, and a printed 0.0 is not restated as a value below 1.0 ppm.

Packaging was analyzed separately. One moist-wipes package was reported at 95.8 ppm antimony, and one breast-wipes package at 38.0 ppm antimony, 0.6 ppm cadmium, 7.6 ppm chromium, and 30.8 ppm lead.[1] Those results describe the packages.

3.2 Korea: digested sheets in micrograms per liter

Choi, Song, and Kim digested the nonwoven sheet of 24 wet wipes, three of them bought in the United States and the rest in Korea, in nitric and hydrofluoric acid by microwave, and measured arsenic, cadmium, lead, nickel, antimony, and thallium by ICP-MS, with mercury measured separately on a direct analyzer without digestion.[2] Each result is the mean of three preparations, reported in micrograms per liter of the digest solution. A dash in their tables means not detected relative to the limits in their Table 4.

Antimony was detected in 12 of the 24 sheets, from 187.86 to 19,558.35 µg/L, and the highest result belongs to the infant sheet labeled rayon alone (code ML). Lead was detected in six sheets, from 7.23 to 55.67 µg/L. Arsenic was detected only in the three United States sheets, from 47.14 to 71.75 µg/L. Nickel was detected in two sheets: 261.26 ± 5.14 µg/L in an infant sheet labeled rayon, polyester, and cotton, and 1,242.63 ± 43.71 µg/L in a United States sheet labeled cloth-like. The other twenty-two sheets had no detectable nickel. Cadmium, mercury, and thallium were not detected in any wet wipe.[2] The mercury quantification limit is internally inconsistent in the paper, printed as 0.80 µg/L in Table 4 and 0.08 µg/L in the methods paragraph, so the mercury non-detects carry that uncertainty.

A content estimate requires a mass and a final volume, and the paper does not supply both unambiguously. Section 2.2 gives approximately 0.2 g of sample in 7 mL of 70 percent nitric acid and 1 mL of 50 percent

hydrofluoric acid, after which the digest was "transferred and diluted with 25 mL" of water; it does not say the solution was made up to 25 mL.[2] If the final volume is taken as 25 mL and the mass as 0.200 g, a solution concentration in µg/L is multiplied by 0.125, and the rayon-labeled sheet reconstructs to 2,444.79 mg/kg. If the 8 mL of acid is assumed to remain and the 25 mL of water is added to it, the volume is 33 mL, the factor is 0.165, and the same sheet reconstructs to 3,227.13 mg/kg. Neither figure is printed in the paper. Both are derived values that depend on an assumption the authors did not state, and neither applies to mercury, which was not digested. They are shown only to convey the order of magnitude the printed solution values could imply. They are not used as results, and they are not compared with any limit.

3.3 China: a per-liter antimony reading from an annex image

On 2 April 2026 the China Association for Quality Promotion of Consumer Products Safety published a monitoring report on 21 baby wipes bought online, tested by a national textile and garment testing laboratory.[3] The sample-by-sample results exist only as an annex image. Read from that image, total antimony on a per-liter label, taken as micrograms per liter, is numeric in 20 of the 21 columns, with stable low and high reads of 15.02 and 450.45 µg/L. One column is not detected. The narrative attributes that non-detect to a named product, which is not named here, and because the fiber cell for that column did not read cleanly, the non-detect is not treated as a result for any fiber type. Lead, cadmium, arsenic, and mercury read as not detected, on a row whose unit could not be read cleanly enough to adopt, and no column for nickel, tin, aluminum, or chromium was read.

The report does not describe the preparation that produced the per-liter antimony figure. It is not the Korean digest, it is not a milligram-per-kilogram content, and the two per-liter datasets are therefore not averaged or compared as if they shared a method. The report states that the Chinese infant-wipe standards it consulted set no antimony limit, and that total antimony was added to the monitoring for that reason.[3]

3.4 Turkey: lotion, fiber, and packaging in parts per billion

Göçer's 2025 master's thesis from İstanbul Üniversitesi-Cerrahpaşa examined ten baby wipes bought in Turkey and coded A through J, and it is the only dataset that separates the three objects that make up a wipe.[4] The liquid was squeezed from the sheet by hand and was not digested. Fiber and packaging were dried, about 0.2 g of each was digested in 65 percent nitric acid, and 10 mL of water was added. The acid volume is not stated, and the thesis does not say the digest was made up to 10 mL, so no milligram-per-kilogram figure can be calculated. One methods sentence reports results in µg/mL, but the calibration, the limits, and the discussion all use ppb of the instrument solution, and the ppb figures are the ones reported here, without rescaling. The calibration extends to 50 ppb.

Antimony was the largest signal. Solution antimony in product I was 695.40 ppb. Fiber antimony was printed as 565.46 ppb (D) and 537.86 ppb (J), and section 4.1 also prints 476.69 ppb (I). All of those antimony results sit above the top of the stated calibration curve. Packaging antimony was 0.03 and 0.04 ppb, at or below the thesis quantification limit of 0.10 ppb.[4]

The other metals were measured at far lower levels, and several maxima are printed differently in different sentences. Solution nickel is given as 53.40 ppb (G) and, elsewhere, as 20.20 ppb with no code; fiber nickel as 28.46 ppb (I) and, elsewhere, 10.53 ppb (B), each described as the highest; packaging nickel was 3.60 ppb (B). Total arsenic was 11.80 ppb in solution A, 1.01 ppb in packaging C, and 0.34 ppb in fiber D. Solution lead is given as 3.10 ppb in section 4.1, fiber lead as 3.11 ppb (F) and 0.29 ppb (J), and packaging lead as 3.20 ppb (H) and 3.16 ppb (J), although a later sentence prints 3.10 and 0.50 ppb as packaging lead for D and C. Solution cadmium was 0.20, 0.20, and 0.30 ppb in A, B, and C, with a packaging maximum of 0.13 ppb (E); fiber cadmium of 0.08 ppb (D) is described as above the quantification limit, although Table 4 sets that limit at

0.11. Total mercury was 2.90 ppb in solution C and 0.63 and 0.67 ppb in fibers A and B; Table 8 prints 2 ppb for solution D, which is not used as the result. Thallium is printed only for fiber, 0.70 ppb in J, while Table 8 describes thallium as below the quantification limit.[4] Where the thesis prints two values for one result, both are kept and neither is averaged away. Chromium appears in the reagent list and not in the results. The thesis's comparisons of its results with drinking-water maxima and with its own restatement of a REACH lead figure are not adopted, because a drinking-water maximum is not a dermal limit and the restated REACH figure does not carry the unit of the legal text.

3.5 Adjacent products: disposable diapers

Two diaper studies describe metals in products worn against the same skin. They are context for the dermal route and are not wipe results.

Nyamukamba and colleagues extracted 20 disposable diapers bought in Cape Town in artificial sweat and in a laboratory urine.[10] In the results text, sweat-extract nickel ranged from 0.022 to 0.197 mg/kg, one urine extract reached 0.224 mg/kg nickel, and sweat-extract total chromium ranged from 0.013 to 0.119 mg/kg. Section 2.3 says the filtrate was analyzed by ICP-MS, whereas the instrument description names an optical emission instrument; the ranges above are taken from the results text rather than from either method sentence. The authors compared their nickel results with an Oeko-Tex value they state as 0.1 mg/kg. The OEKO-TEX STANDARD 100 Annex 4 edition held by the Index, edition 02.2025, lists class I extractable nickel at 1.0 mg/kg.[11] Because the two figures differ, and because a diaper extract is not a wipe, this synthesis does not score these diaper results against either figure. The panel did not include tin, aluminum, or mercury.

Šmajgl and Obhođaš measured total tin by energy-dispersive X-ray fluorescence in disposable diaper parts.[12] As reported in the publisher abstract, tin in diaper top sheets ranged from below 1.5 to 23.1 mg/kg and in adhesive tape from below 1.5 to 22.4 mg/kg. The authors suggest an organotin catalyst or a PVC heat stabilizer as the origin, but the method measures total tin and does not identify organotin species. These abstract ranges are diaper-part results and are not compared with any limit here.

Table 1. Metal results in baby wipes, by dataset and in each dataset's own unit

No column is converted to another's unit, and no row is a market rate. "Not in panel" means the element was not analyzed. Sources: Washington[1]; Korea[2]; China[3]; Turkey[4].

METAL	WASHINGTON COMPONENTS (PPM)	KOREAN SHEET DIGESTS (µg/L)	CHINESE ANNEX (µg/L)	TURKISH THESIS (INSTRUMENT PPB)	SPECIES OR ENDPOINT NOT MEASURED
Antimony	40.6, 61.3, 70.6	Detected in 12 of 24; 187.86 to 19,558.35; highest labeled rayon	Numeric in 20 of 21; 15.02 to 450.45	Solution 695.40; fiber 565.46, 537.86, and 476.69; packaging 0.03 and 0.04	Oxidation state

Lead	0.4, 0.8, 1.4	Detected in six; 7.23 to 55.67	Not detected; row unit not adopted	Solution 3.10; fiber 3.11 and 0.29; packaging 3.20 and 3.16, with a conflicting sentence	None beyond the total
Arsenic, total	0.0 in all three	Detected in the three United States sheets; 47.14 to 71.75	Not detected; row unit not adopted	Solution 11.80; fiber 0.34; packaging 1.01	Inorganic arsenic
Mercury, total	0.0	Not detected	Not detected; row unit not adopted	Solution 2.90; fiber 0.63 and 0.67	Methylmercury
Cadmium	0.0	Not detected	Not detected; row unit not adopted	Solution 0.20, 0.20, and 0.30; fiber 0.08, below the stated quantification limit; packaging 0.13	None beyond the total
Chromium, total	0.5 to 0.6	Not in panel	Not read	Not in results	Hexavalent chromium
Nickel	Not in panel	Detected in two; 261.26 and 1,242.63	Not read	Solution 53.40 or 20.20; fiber 28.46 or 10.53; packaging 3.60	Release per unit area
Tin	Not in panel	Not in panel	Not read	Not in panel	Total tin and organotin species
Aluminum	Not in panel	Not in panel	Not read	Not in panel	All forms

4 | 4. Antimony and the polyester-catalyst hypothesis

Antimony is the one metal detected in every wipe dataset, and the most direct explanation for it is the chemistry of polyester. Choi, Song, and Kim attribute the high antimony in their nonwovens to antimony compounds used as a PET catalyst.[2] Textile research supports the general mechanism: Biver, Turner, and Filella measured total antimony of about 125 to 470 µg/g in polyester textiles intended for skin contact, with Table 1 listing 125, 185, 286, 357, 452, and 471 µg/g.[13] Manufacturer filings to United States state chemical-reporting systems point the same way. The one filing that concerns a wipe and reports a metal, for a reusable cotton wipe that its maker filed under baby hygiene categories rather than the personal-wipes category, reports antimony as a catalyst in a synthetic-fiber component.[14]

The Korean table does not turn that mechanism into a rule for wipes. Its highest antimony result is the sheet labeled rayon alone, several sheets labeled rayon plus polyester or rayon plus PET had no detectable antimony, and the pulp-with-polypropylene and cotton-labeled sheets had none.[2] A high antimony result on a rayon label can mean that the label is incomplete, that the antimony has another source, or both, and the published data cannot distinguish among those readings. The Washington report does not name the fiber of its three

components,[1] and the Turkish thesis does not identify fiber type for its ten products, attributing its high antimony instead to a flame retardant, a hypothesis it does not test by measurement.[4] The catalyst account is therefore the right hypothesis to test on polyester sheets, with fiber identity established by analysis and antimony measured on the same sheet. It is not, on current evidence, a finding about any particular wipe.

5 | 5. Hazard by metal, and what is known on infant skin

Each metal's hazard is described here by the endpoint that usually drives its assessment, followed by what is known about that metal in a wipe or on infant skin. A classification or reference value derived from ingestion or inhalation is not a dermal dose from a wipe.

Infant skin is the exposure surface, and no study in the corpus quantifies the factors that make it distinctive for wipe use: stratum corneum thickness, contact time in the diaper area, or the ratio of skin area to body weight. The reasons those factors matter are well understood in general terms. The barrier is immature in early infancy, the diaper area is occluded and often wet or already inflamed, and an infant's skin area is large relative to its body weight, while occlusion, maceration, and dermatitis can each change absorption. These are reasons to measure the dermal route. They are not a measured fraction, and no study reports the share of any metal that crosses intact or dermatitic infant skin from a wipe sheet or a wipe lotion.

Lead

Lead is a developmental neurotoxicant, and effects on the developing nervous system dominate infant and child assessments. The wipe evidence is a concentration rather than a dose: 0.4 to 1.4 ppm in three screened Washington components and 7.23 to 55.67 µg/L in six Korean digests.[1,2] Neither study measured blood lead, dermal absorption, or hand-to-mouth transfer, and the lead in one Washington package describes the package rather than the wipe.

Arsenic

Regulatory food work isolates inorganic arsenic as the hazardous form, with carcinogenicity and, at lower exposures, developmental and dermal effects attaching to the inorganic species rather than to every total-arsenic figure. Every wipe study reports the element only. The three Korean detects, 47.14 to 71.75 µg/L of total arsenic in the United States sheets, are a reason to speciate, not an inorganic-arsenic result.[2] Inorganic arsenic in a baby wipe has not been measured.

Mercury

Methylmercury is the organic species associated with neurodevelopmental harm after ingestion, while elemental and inorganic mercury differ in their kinetics and include dermal and inhalation routes for some forms. The wipe measurements are total mercury. Washington prints 0.0 and the Korean study detected none,[1,2] but a non-detect on a total-mercury method does not establish absence below the method limit, and the Korean quantification limit is itself inconsistent. The Turkish thesis reports total mercury in its solution and fiber fractions, in instrument ppb (Table 1).[4] Methylmercury in a baby wipe has not been measured.

Cadmium

Cadmium's critical long-term endpoint in dietary assessments is kidney tubular dysfunction, with bone effects at related exposures, and that record is oral and cumulative. Cadmium was printed as 0.0 in the Washington components and not detected in any Korean sheet.[1,2] A printed zero in one small panel and a non-detect in

 © All rights reserved

another do not show that every sheet and lotion is free of cadmium, and no dermal-absorption fraction for cadmium from a wipe has been measured.

Chromium

Chromium's toxicity depends on oxidation state. Trivalent chromium is an essential nutrient at low doses and is poorly absorbed by mouth, whereas hexavalent chromium is the sensitizer and the carcinogenic species in an occupational record that is largely inhalation and, for sensitization, skin. Washington measured total chromium at 0.5 to 0.6 ppm and stated that hexavalent chromium was outside the study, and the Korean panel did not include chromium.[1,2] A total-chromium result cannot be read as hexavalent chromium or as a sensitization test, and hexavalent chromium in a baby wipe has not been measured.

Nickel

Nickel is the most common cause of metal allergic contact dermatitis. Sensitization is acquired through the skin, and once a person is sensitized, later contact can elicit a rash. The clinical literature distinguishes a sensitization threshold from an elicitation threshold, and both are usually expressed as release per unit area over time, not as a bulk concentration in a wet sheet.

The nearest population figure is a prevalence among children who were patch tested, not a threshold. Isufi and colleagues, in a systematic review and meta-analysis of patch-tested children under 18 covering 2010 to 2024, found nickel sulphate positive in 11.9 percent (95 percent confidence interval 8.6 to 15.71) across 16 studies, the highest prevalence of any allergen in that selected population.[15] Those children were referred for testing, not sampled from the general population, and they were not tested because of wipes.

European REACH Annex XVII entry 27 limits nickel release from piercing posts to below 0.2 µg/cm²/week and from articles in prolonged skin contact, such as jewelry and garment hardware, to 0.5 µg/cm²/week.[16] A wet wipe is not one of those articles, and a release rule for metal objects is not an elicitation threshold measured on infant skin under a wipe. No study reviewed here shows that nickel in a baby wipe caused diaper dermatitis, and the published reports of wipe-associated allergic contact dermatitis identified for this synthesis implicate preservatives and fragrance rather than metals. The two Korean nickel detects remain solution concentrations and have not been converted to a content or to a release per square centimeter.[2]

Tin

Inorganic tin in food is an acute gastric irritant at intakes far above ordinary trace levels, and that record concerns canned food rather than skin. Organotin compounds are a distinct hazard class, used as biocides, PVC stabilizers, and some polymerization catalysts. The diaper study that found total tin proposes an organotin catalyst or a PVC stabilizer as its origin without speciating it.[12] No measurement of tin in any form has been reported for a baby wipe.

Aluminum

Aluminum's neurotoxicity and bone toxicity are documented at high systemic exposures, particularly when the kidney cannot clear the metal, and that record does not transfer to a wipe by analogy. Aluminum has not been measured in a baby wipe, a wipe lotion, or a wipe nonwoven, and the lotion formulation described by Cheriaa and Boubaker names no aluminum salt.[5] Papermaking alum and aluminum packaging are conceivable sources that no wipe study has tested.

Antimony

The International Agency for Research on Cancer, in Monograph Volume 131 (2023), classified trivalent antimony as probably carcinogenic to humans (Group 2A), with antimony trioxide among the trivalent compounds evaluated.[17] The Group 2A rationale is limited evidence for cancer in humans, drawn from lung cancer in occupational cohorts, sufficient evidence in experimental animals, and strong mechanistic evidence. Pentavalent antimony was evaluated as Group 3, not classifiable, and the volume notes that antimony trioxide alone had been evaluated as possibly carcinogenic (Group 2B) in 1989. The monograph describes inhalation as the significant occupational route and names dermal contact with dust as a concern in production and processing, including a modelled textile flame-retardant scenario. None of that record concerns an infant, a wipe, or a dermal dose from a lotion.[17]

The wipe measurements are total antimony. They do not identify the oxide, the oxidation state, or the amount that leaves the fiber into sweat or lotion. In the polyester-textile experiments of Biver, Turner, and Filella, default artificial sweat mobilized about 0.05 to 2 percent of total antimony, 0.1 to 1 µg/g in the abstract, although Table 1 includes an extractable result of 2.572 µg/g.[13] Those were clothing and household textiles under standard sweat tests, not wipes on diaper-area skin, and the fraction is not applied to wipes here. A Group 2A classification is a hazard classification for the trivalent agent. It is not a finding that a baby wipe causes cancer, and this synthesis makes no such finding.

6 | 6. Why a concentration in a wipe is not yet a dose

A dermal dose from a wipe has at least four inputs: the metal content of the sheet and the lotion, the amount of product that touches skin, the time it stays there, and the fraction absorbed. Frequency of use and contact time are real variables, but no time-and-motion or diary study of baby-wipe use is in the corpus, so they are not assigned numbers here.

An illustrative dose is deliberately not offered. The reconstructed Korean content figures are estimates of what a digest implies about a sheet under an unstated volume assumption, not doses. The approximately 0.2 g in that study is the mass placed in a digestion vessel, not the mass of a wipe a caregiver uses, and the 25 mL is water added to the digest, not a wipe's lotion load.[2] No source supplies together a wipe mass, a use frequency, a contact time, and a dermal absorption fraction for any metal. Filling those gaps with assumptions would produce a number that looks like a dose without being one, so the absence of those inputs is reported as the finding.

What can be said without arithmetic is qualitative and still useful. Antimony is present in wipes and wipe parts across four datasets, on bases that cannot yet be compared with one another or with a cosmetic limit. Nickel is present in two digested sheets and undetected, at the method's limit, in the other twenty-two. Lead is a low-ppm signal in three screened components and a µg/L signal in six digests. The dermal step is unmeasured for every metal.

7 | 7. The regulatory landscape

A baby wipe is not one legal object. A product applied to the body for cleansing can fall within a cosmetic definition, a children's product can fall within state reporting laws, and a voluntary textile certification is neither. The instruments below are described in their own units and with their own legal character, and the wipe datasets are not scored against them: the Washington preparation is undescribed and its samples predate the later statutes, while the other three datasets report solution concentrations rather than content.

The European Union Cosmetics Regulation, Regulation (EC) No 1223/2009, prohibits in Annex II the use as ingredients of arsenic, antimony, cadmium, chromium and its salts, lead, mercury (with an Annex V exception), and elemental nickel, with many nickel salts listed as separate entries.[18] Article 17 allows technically unavoidable traces and sets no ppm figure.

Korea's cosmetic safety standard, Article 6(2), as reproduced by the Korea Cosmetic Industry Association, applies where a substance was not intentionally added and complete removal is technically impossible.[19] It sets lead at or below 50 µg/g in clay powders and 20 µg/g in other products; nickel at or below 35 µg/g in eye makeup, 30 µg/g in color makeup, and 10 µg/g in other products; arsenic at or below 10 µg/g; mercury at or below 1 µg/g; antimony at or below 10 µg/g; and cadmium at or below 5 µg/g. The same article names wet wipes for methanol, formaldehyde, and microbial limits, so wet wipes fall within the instrument, but it gives no wet-wipe-specific figure for any metal, and tin, aluminum, and chromium are not in its impurity list.

Health Canada's guidance on heavy-metal impurities in cosmetics, section 4, treats lead above 10 ppm, arsenic above 3 ppm, cadmium above 3 ppm, mercury above 1 ppm, and antimony above 5 ppm as technically avoidable.[20] Nickel does not appear, the guidance is not a regulation, and it does not name wipes.

The German Federal Office of Consumer Protection and Food Safety (BVL) published orientation values in 2017 of 2.0 mg/kg lead, 0.1 mg/kg cadmium, 0.1 mg/kg mercury, 0.5 mg/kg arsenic, and 0.5 mg/kg antimony for cosmetics in general, with no nickel line.[21] The monitoring set behind those values was decorative cosmetics, baby powder, and toothpaste, not wipes, and orientation values are not a statute.

Washington's RCW 70A.560.020(2) provides that from 1 January 2025 a cosmetic may not be sold if it contains intentionally added lead or lead at 1 ppm or above.[22] One ppm is 1,000 µg/kg. The cosmetic definition used there does not name wipes, and the Washington wipe rows are not treated as violations, because their basis may not be the whole article and the samples predate the statute. In the United States, 21 CFR 700.13 restricts mercury in cosmetics to an unavoidable trace below 1 ppm, with a higher allowance for certain eye-area products.[23]

REACH Annex XVII entry 20 restricts specified organotin compounds above 0.1 percent by weight of tin: tri-substituted organotins in articles, dibutyltin in mixtures and articles for the general public, and dioctyltin in a named list that includes textile articles intended for skin contact, childcare articles, female hygiene products, and nappies.[24] A baby wipe is not named, and the entry is a species rule rather than a trace limit for total tin.

OEKO-TEX STANDARD 100 is a private certification. Its Annex 4, edition 02.2025, sets class I extractable limits including arsenic 0.2, cadmium 0.1, chromium 1.0, chromium (VI) 0.5, mercury 0.02, nickel 1.0, lead 0.2, and antimony 30.0 mg/kg.[11] Extractable content and total content are different tests, so an extractable limit is not a score for a total digest. The standard's Annex 6 is an optional catalogue, distinct from Annex 4, and neither is law.

China's GB/T 27728.2-2024 is a recommended standard for wet wipes for children. A laboratory summary of the standard lists lead, arsenic, cadmium, and mercury as test items for infant wipes and lists neither nickel nor antimony; the summary gives no numeric limits, and the standard's own text was not available.[25]

Manufacturer reporting under United States state children's-product laws is disclosure, not limitation. Filings in the personal-wipes category of the shared state reporting database list preservatives and a phthalate, and none reports a metal.[14]

Table 2. Metal figures in instruments that could be argued to reach a baby wipe

Each figure is shown in its instrument's own unit and legal character. None is a score for the wipe datasets in Table 1, and "none" means the instrument sets no figure for that metal. Sources: EU[18]; Korea[19]; Health Canada[20]; BVL[21]; Washington[22]; United States[23]; OEKO-TEX[11].

INSTRUMENT	LEGAL CHARACTER	BASIS	LEAD	ARSENIC	MERCURY	CADMIUM	NICKEL	ANTIMONY
EU Regulation (EC) No 1223/2009	Regulation	Ingredient prohibition; trace allowance without a figure	None	None	None	None	None	None
Korea safety standard, Article 6(2)	Government standard, for unintentional traces	Content, µg/g	20 (50 in clay powders)	10	1	5	10 (30 color makeup, 35 eye makeup)	10
Health Canada guidance, section 4	Guidance	Content, ppm	10	3	1	3	None	5
BVL 2017 orientation values	Orientation value	Content, mg/kg, cosmetics in general	2.0	0.5	0.1	0.1	None	0.5
Washington RCW 70A.560.020(2)	State statute	Content, ppm	1	None	None	None	None	None
21 CFR 700.13	Federal regulation	Content, ppm	None	None	Below 1	None	None	None
OEKO-TEX STANDARD 100, Annex 4, class I	Private certification	Extractable, mg/kg	0.2	0.2	0.02	0.1	1.0	30.0

Antimony is the difficult case. It is the metal the wipe studies detect most readily, and several impurity lists include it, Korea at 10 µg/g, Health Canada guidance at 5 ppm, and the BVL orientation value at 0.5 mg/kg, yet no wipe dataset is on a basis that can be compared with those figures. A regulator reading only food-metal panels would miss it, because antimony is not among the four metals those panels usually lead with. Nickel is difficult in a different way: Korea sets a content figure for products other than eye and color makeup, Health Canada does not list it, and the European article rule concerns release from metal objects, so a wipe standard modeled on any one of those instruments would not match the others.

8 | 8. Implications

For parents and caregivers

The evidence supports the following statements. Antimony has been measured in baby-wipe components, in digested wet-wipe sheets, in the expressed lotion and fiber of coded products, and in an annex of 21 online products, on bases too unlike one another to describe the market. Lead appears at low ppm in three screened components and at micrograms per liter in six digests. Nickel appears in two digested sheets and not in the other twenty-two of that study. Tin, aluminum, inorganic arsenic, methylmercury, and hexavalent chromium have not been measured in a wipe or have not been speciated, which is different from having been shown to be absent.

It does not support a risk verdict. No study follows a wipe from the package onto infant skin and into the body, and whether to use or stop using a product is a caregiving decision this synthesis does not make.

For manufacturers

The sheet and the liquid need separate results in a content unit that a limit can be compared with. The Turkish thesis shows that the split is practical, and a single number for "the wipe" cannot be assigned to the water, the preservative system, or the fiber. Fiber identity should be recorded by analysis as well as by label, because a polyester label does not prove antimony is present and a rayon label does not prove it is absent. If a claim concerns antimony trioxide, the species should be identified. The package needs its own row, both because it can carry metal of its own and because a package left in the same digest as the sheet would confound the result. A total digest is not a sensitization test: a laboratory that wants to speak to allergic contact dermatitis has to measure release into sweat or a simulant over a stated time and area.

For regulators

The legal character of each number should remain visible. A Korean impurity maximum, a Canadian guidance figure, a German orientation value, a Washington lead statute, and a private extractable textile limit are not interchangeable, and none was written as a score for a digest reported in micrograms per liter. Where no government sets a number for a metal in this product, the gap should stay a gap; borrowing a jewelry release limit or a diaper criterion to fill tin, aluminum, or hexavalent chromium would present a policy choice as if it were a measurement. Speciation rules already applied to food should not be relaxed because the matrix is a wipe: total arsenic is not inorganic arsenic, total mercury is not methylmercury, total chromium is not hexavalent chromium, and total tin is not an organotin.

For researchers

The next useful study is small and specific. It would digest or extract the sheet and the lotion separately, report mass, final volume, and the result in mg/kg of each fraction, and keep the solution concentration as a method detail rather than the reported unit. It would determine fiber identity analytically; include tin, aluminum, and chromium; speciate arsenic, mercury, chromium, and any tin found; and identify the antimony species if the trioxide is discussed. It would run a sweat or simulant release alongside total content, because sensitization questions are release questions, and it would sample without a screening trigger, so that the results can describe a market rather than a list of screened positives.

9 | 9. Limitations

The measured record is small and non-random. The Washington rows were selected because a screen had already found a metal, the Korean set is a convenience sample of 24 sheets, the Chinese results are optical reads of an annex image with no described preparation, and the Turkish results come from a thesis that has not been through journal peer review, with several internally conflicting maxima and antimony results above its calibration range. None of these datasets reports a content in mg/kg for the sheet and the lotion separately, and none

measures absorption. Several supporting documents were available only in partial form: the diaper-tin ranges come from the publisher abstract, the lotion formulation from a publisher extract, the Korean standard from an association reproduction, the Canadian guidance from an archived capture, and the Chinese national standard from a laboratory summary. Non-metal constituents of wipes, including preservatives, fragrance, phthalates, microplastic fibers, and per- and polyfluoroalkyl substances, are outside the scope of this synthesis and are the subject of separate work.

10 | 10. Conclusion

The heavy-metal record for baby wipes supports a narrow set of conclusions. Antimony is the metal the measurements find, in every dataset and in both the sheet and the expressed lotion, and polyester catalyst chemistry is the leading hypothesis for its origin, although the published fiber labels do not yet confirm it. Lead, arsenic, and nickel appear intermittently at levels that cannot be placed on a common basis, and cadmium and mercury are printed as 0.0 or not detected in three of the four datasets and appear only at low instrument-solution levels in the fourth. Tin, aluminum, and the hazardous species of arsenic, mercury, and chromium have not been measured at all.

What the record cannot yet support is a statement about dose. The datasets report on four incompatible bases, none of them the content basis that cosmetic limits assume, and no study has measured transfer from a wipe to infant skin or absorption through it. The work that would change that is well defined: separate content results for sheet and lotion, analytical fiber identity, speciation, release testing, and absorption measurement on intact and compromised skin. Until that work is done, a concentration in a baby wipe remains a concentration in a baby wipe, and not yet a dose to an infant.

11 | Anchor sources

These Heavy Metal Index source records carry the measurements this synthesis rests on. Each holds the ingest receipt, the values extracted from the document, and the file hash of the document it was built from.

- stone2021-wa-metals-childrens-consumer-products: Washington State laboratory table of metals in three screened wipe components and two wipe packages, in ppm.
- choi2016-wet-wipes-nonwoven-heavy-metals: Metals in microwave digests of 24 wet-wipe nonwoven sheets, in µg/L of digest solution, with the authors' fiber labels.
- cpqs2026-online-baby-wipes-quality-monitoring: Per-liter total antimony read from the annex image of a 2026 monitoring report on 21 online baby wipes.
- gocer2025-baby-wet-wipes-solution-fiber-packaging: Metals in the expressed lotion, digested fiber, and digested packaging of ten coded baby wipes, in instrument ppb.
- biver2021-antimony-release-polyester-artificial-sweat: Total and sweat-extractable antimony in polyester textiles; the mechanism and release evidence for the catalyst hypothesis.
- nyamukamba2023-south-africa-diapers-metals-formaldehyde: Sweat- and urine-extractable metals in disposable diapers; adjacent-product context for the dermal route.
- smajgl2015-tin-baby-diapers-croatia: Total tin in disposable diaper parts by X-ray fluorescence; adjacent-product context for tin and organotin.

 © All rights reserved

12 | How this page was published

This synthesis was established on 25 September 2026 from the Heavy Metal Index evidence set for baby wipes and later rewritten in full, with the evidence base unchanged. Under the Heavy Metal Index model it is a preprint until external review is recorded below.

13 | How to cite

Pendergrass, K. (2026). Heavy metals in baby wipes. Institute of Contaminant Standards.

References

[1] Metals in Children's and Consumer Products and Packaging
Alex Stone · Washington State Department of Ecology, Publication 14-04-014 · 2014, contact revision 2021 · apps.ecology.wa.gov
GOVERNMENT REPORT

[2] Determination of heavy metals on the non-woven in wet wipes using ICP-MS
Sung-Min Choi, Jin-Kun Song, and Sang-Jin Kim · Journal of the Korean Oil Chemists' Society · 2016 · doi.org/10.12925/jkocs.2016.33.1.195
PEER-REVIEWED

[3] Analysis report: quality monitoring results for baby wet wipes sold online
China Association for Quality Promotion of Consumer Products Safety · 2 April 2026 · cpqs.org.cn/archives/3229
ASSOCIATION REPORT

[4] Evaluation of the liquid, fiber and packaging content of baby wet wipes for heavy metal contamination
İlker Aras Göçer · MSc thesis, İstanbul Üniversitesi-Cerrahpaşa · 2025 · YÖK thesis catalog 988276
THESIS

[5] Quality assessment of baby wet wipes
Rim Cheriaa and Jaouachi Boubaker · Journal of Industrial Textiles 51(2_suppl):2124S-2147S · 2022, online 9 December 2021 · doi.org/10.1177/15280837211046619
PEER-REVIEWED

[6] Quality Assurance Project Plan: Parabens and Metals in Children's Cosmetic and Personal Care Products
Alex Stone · Washington State Department of Ecology, Publication 12-07-021 · 2012
GOVERNMENT PLAN

[7] Quality Assurance Project Plan: Phthalates and Metals in Packaging
Alex Stone · Washington State Department of Ecology, Publication 12-07-022 · 2012
GOVERNMENT PLAN

[8] Quality Assurance Project Plan: Phthalates and Metals in Children's Products
Alex Stone · Washington State Department of Ecology, Publication 12-07-023 · 2012
GOVERNMENT PLAN

[9] Quality Assurance Project Plan: Formaldehyde, Volatile Organic Compounds and Metals in Children's Products
Alex Stone · Washington State Department of Ecology, Publication 12-07-024 · 2012
GOVERNMENT PLAN

[10] Determination of selected harmful substances in baby diapers available on the South African market
Nyamukamba P, Mququ Z, Nkosi S, and Isaacs S · International Journal of Environmental Research and Public Health · 2023 · doi.org/10.3390/ijerph20021023
PEER-REVIEWED

[11] OEKO-TEX STANDARD 100: limit values and individual substances according to Appendices 4 and 5
OEKO-TEX · Annex 4, edition 02.2025 · 2025 · oeko-tex.com
PRIVATE STANDARD

 © All rights reserved

[12] Occurrence of tin in disposable baby diapers
D. Šmajgl and J. Obhođaš · X-Ray Spectrometry · 2015 · doi.org/10.1002/xrs.2609
PEER-REVIEWED

[13] Antimony release from polyester textiles by artificial sweat solutions: A call for a standardized procedure
Marc Biver, Andrew Turner, and Montserrat Filella · Regulatory Toxicology and Pharmacology · 2021 · doi.org/10.1016/j.yrtph.2020.104824
PEER-REVIEWED

[14] High Priority Chemicals Data System
Interstate Chemicals Clearinghouse · public reporting system for the Washington, Oregon, and Vermont children's-product laws, product category Wipes - Personal, queried 25 September 2026 · hpcds.theic2.org
GOVERNMENT DATABASE

[15] Allergens responsible for contact allergy in children from 2010 to 2024: A systematic review and meta-analysis
Daniel Isufi, Mikkel Bak Jensen, Christoffer Kursawe Larsen, Farzad Alinaghi, Jakob Ferløv Baselius Schwensen, and Jeanne D. Johansen · Contact Dermatitis 92:327-343 · 2025 · doi.org/10.1111/cod.14753
PEER-REVIEWED

[16] REACH Annex XVII: restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles
CIRS C&K Testing · consolidated reproduction of REACH Annex XVII · 2023
REGULATORY TEXT

[17] IARC Monographs Volume 131, trivalent and pentavalent antimony
International Agency for Research on Cancer · 2023 · NCBI Bookshelf NBK598316
MONOGRAPH

[18] Regulation (EC) No 1223/2009 on cosmetic products
European Parliament and Council of the European Union · Annex II and Article 17 · 2009 · eur-lex.europa.eu
REGULATORY TEXT

[19] Korea cosmetic safety standard, Article 6(2)
Ministry of Food and Drug Safety, association reproduction opened 25 September 2026 · kcia.or.kr
REGULATORY TEXT

[20] Guidance on heavy metal impurities in cosmetics
Health Canada · section 4, Internet Archive capture of 10 January 2025 · canada.ca
GUIDANCE

[21] Technically avoidable heavy metal contents in cosmetic products
Federal Office of Consumer Protection and Food Safety (BVL) · Journal of Consumer Protection and Food Safety 12:51-53 · 2017, online 6 October 2016 · doi.org/10.1007/s00003-016-1044-2
PEER-REVIEWED

[22] Revised Code of Washington 70A.560.020
Washington State Legislature · subsection (2), lead in cosmetic products · app.leg.wa.gov
STATUTE

[23] Code of Federal Regulations, Title 21, section 700.13
U.S. Food and Drug Administration · mercury in cosmetics · ecfr.gov
REGULATION

[24] EU REACH Annex XVII entry 20: organotin compounds in skin-contact and childcare articles
European Commission · REACH restriction
REGULATORY TEXT

[25] GB/T 27728.2-2024, wet wipes for children: test-item summary
Intertek China · summary of the national wet-wipes standard series · intertek.com.cn
STANDARD SUMMARY

 © All rights reserved

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this finding. Published as an institutional technical report with its review state tracked on the article page. © All rights reserved. Institute of Contaminant Standards / Heavy Metal Index.

DATA AVAILABILITY

Every source behind this finding is listed with its own extraction record at https://heavymetalindex.com/synthesis/heavy-metals-in-baby-wipes#sources, each linking to the ingest receipt, the extracted values, and the file hash of the document it was built from.

COMPETING INTERESTS

The Heavy Metal Index reports what the peer-reviewed and regulatory literature supports. It is operated separately from any certification program, does not evaluate or endorse brands, and does not use certification thresholds as evidence for the statements made here.

PUBLICATION HISTORY

First published 2026-09-25. This revision 2026-09-26. The full dated record is maintained at https://heavymetalindex.com/synthesis/heavy-metals-in-baby-wipes.